Huawei Access Node MALL-ACC-03 QoS Congestion Event Troubleshooting SOP

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4364

Description

The failure domain of QoS congestion events, specifically microbursts, interface tail drops, and QoS queue starvation on Huawei Access nodes, occurs due to the intricate interplay between the node's queue scheduling algorithms, interface buffer sizes, and the prioritization of traffic flows. The Huawei Access node's QoS architecture is based on the Multi-Field Classification, Label Switching, and Traffic Policing mechanisms. In the event of a congestion event, the node's queue scheduling algorithms, such as Weighted Random Early Detection (WRED) and Priority Queuing (PQ), may not be able to effectively manage the traffic, leading to microbursts, interface tail drops, and QoS queue starvation. Furthermore, the node's interface buffer sizes and the configuration of the QoS policies may also contribute to the congestion event. Therefore, it is essential to perform a thorough analysis of the node's QoS configuration, traffic patterns, and hardware resources to identify the root cause of the congestion event and implement the necessary corrective actions.

Pre-requisite Checks

- Perform a site visit to the Malleswaram Node and check the physical and environmental conditions.

- Verify the status of the redundant link and ensure it is not causing the congestion event.

- Check the node's software version and ensure it is up-to-date.

- Perform a thorough review of the node's QoS configuration and verify that it is correctly configured to manage the traffic flows.

Potential Root Causes

Cause	Probability
Inadequate QoS configuration	High
Insufficient interface buffer sizes	Medium
Hardware resource constraints	Low

Diagnostic Commands & Expected Outputs

```
Command: display queueing queue , Expected Output: Queueing information for queue ., Description: This command displays the queueing information for the specified queue, including the number of packets in the queue, the number of packets dropped, and the queue's scheduling algorithm.

Command: display interface statistics, Expected Output: Interface statistics for ., Description: This command displays the interface
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statistics for the specified interface, including the number of
packets received, the number of packets transmitted, and the
interface's buffer utilization.

Command: display qos policy , Expected Output: QoS policy information
for ., Description: This command displays the QoS policy information
for the specified policy, including the policy's classification rules,
traffic policing configuration, and queue scheduling algorithms.
```

Resolution Steps

Step 1: Step: Update the QoS configuration to improve traffic management, Command: system-view qos routing 2 class-map match-all match ip address policy-map class dequeue bandwidth , Description: This step updates the QoS configuration to improve traffic management by creating a new class-map, policy-map, and queue, and configuring the dequeue and bandwidth values accordingly.

Step 2: Step: Increase the interface buffer sizes, Command: interface buffer-size , Description: This step increases the interface buffer sizes to improve the node's ability to handle traffic bursts and reduce the likelihood of interface tail drops.

Step 3: Step: Verify the QoS configuration and traffic patterns, Command: display qos routing 2 qos policy , Description: This step verifies the QoS configuration and traffic patterns to ensure that the updates are correctly applied and that the congestion event has been resolved.

Post-Check Verification

```
Command: display queueing queue , Expected Output: Queueing
information for queue , indicating that the queue is stable and not
experiencing drops., Description: This command verifies that the queue
is stable and not experiencing drops, indicating that the congestion
event has been resolved.

Command: display interface statistics, Expected Output: Interface
statistics for , indicating that the interface is not experiencing
buffer utilization above 80%., Description: This command verifies that
the interface is not experiencing buffer utilization above 80%,
indicating that the congestion event has been resolved.

Command: display qos policy , Expected Output: QoS policy information
for , indicating that the policy is correctly configured and traffic
is being managed according to the policy., Description: This command
verifies that the QoS policy is correctly configured and traffic is
being managed according to the policy, indicating that the congestion
event has been resolved.
```

Roll-back Procedures

- Step: Restore the original QoS configuration, Command: system-view qos routing 2 undo qos policy , Description: This step restores the original QoS configuration to prevent any further disruptions to the network.

- Step: Verify the network's stability after rollback, Command: display queueing queue display interface statistics display qos policy , Description: This step verifies the network's stability after rollback by monitoring the queueing information, interface statistics, and QoS policy information.